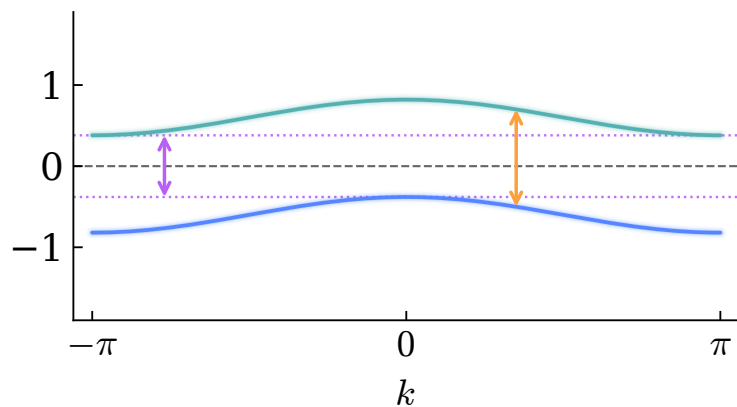


Energy (arbitrary units)

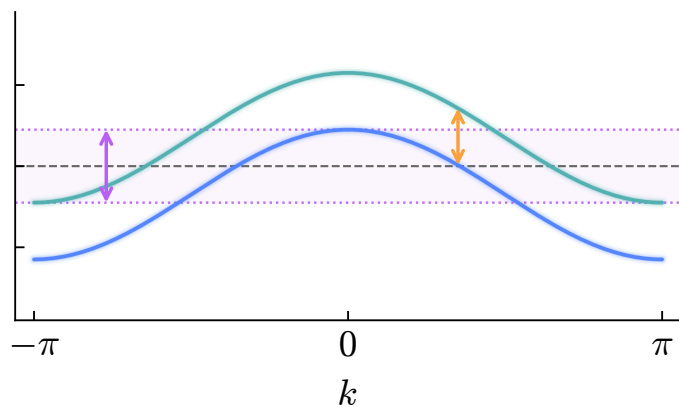
(a) Global gap

$$\Delta_{\text{dir}}^{\text{min}} > 0, \Delta_{\text{ind}} > 0$$



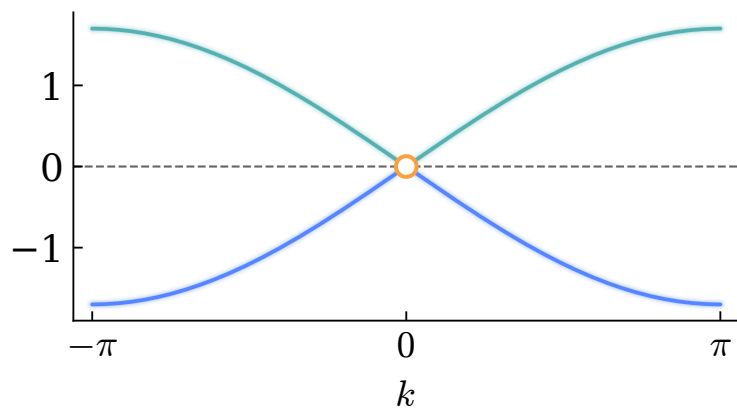
(b) Energy overlap

$$\Delta_{\text{dir}}^{\text{min}} > 0, \Delta_{\text{ind}} < 0$$



(c) Boundary contact

$$\Delta_{\text{dir}}^{\text{min}} = 0$$



(d) Internal crossing

$$\Delta_{\text{dir}}^{\text{min}} > 0$$

